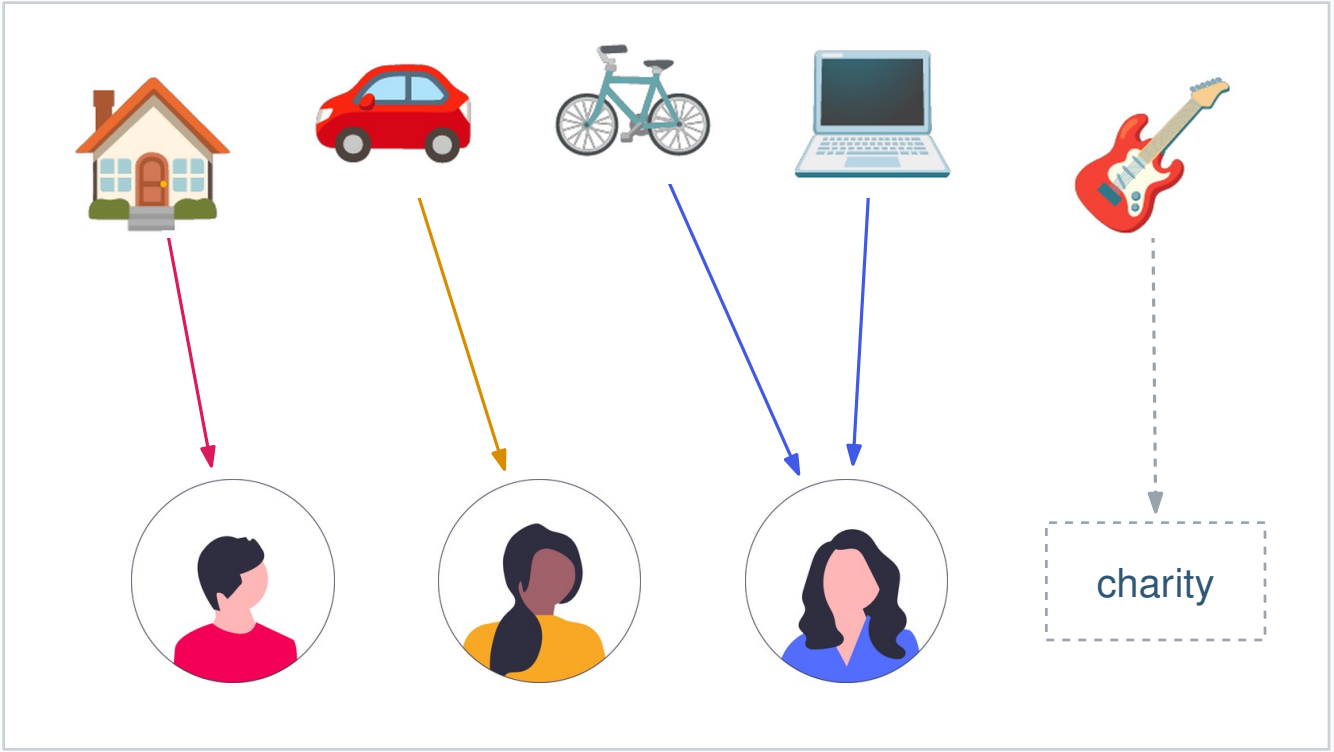
Fair Division

Lecture 05:
EFX with
Charity



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Where We Are

Recall. X is **EFX** if $v_i(X_i) \geq v_i(X_j \setminus \{g\})$ for all agents i, j and **all** $g \in X_j$.

- identical valuations, or two agents: complete EFX allocations **exist** [Plaut, Roughgarden 2018] (end of today)
- complete EFX allocations need **not** exist for monotone (even submodular) valuations [Akrami, Mayorov, Mehlhorn, Srinivas, Weidenbach 2026; Mackenzie, Suzuki 2026]
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Question. Can we donate only a **few** goods that **nobody wants**?

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Lecture 4. An EFX **partial** allocation with $\mathrm{NW} \geq 2^{-(1-1/n)}\,\mathrm{opt}$. But many goods may be donated, and agents may envy the donated goods.

Yes: fewer than n goods, and nobody envies them [Chaudhury, Kavitha, Mehlhorn, Sgouritsa 2020].

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(ii) nobody envies the pool: $v_i(X_i) \geq v_i(P)$ for every agent i;
(iii) $|P|$ is smaller than the number of sources of G_X; in particular $|P| < n$.

Recall: the envy graph G_X has an edge $i \to j$ if i envies j; a **source** is an agent whom nobody envies.

Envy, Strong Envy and Rotations

Definition. Agent i **envies** $S \subseteq M$ if $v_i(S) > v_i(X_i)$; S is **envied** if some agent envies it. Agent i **strongly envies** S if $v_i(S \setminus \{h\}) > v_i(X_i)$ for some $h \in S$.

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Question. Why is X^C EFX?

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Same bundles, new owners. Towards her old bundle B: $v_i(B) = v_i(X_i) \leq v_i(X_i^C)$. Towards another bundle B: $v_i(B \setminus \{g\}) \leq v_i(X_i) \leq v_i(X_i^C)$.

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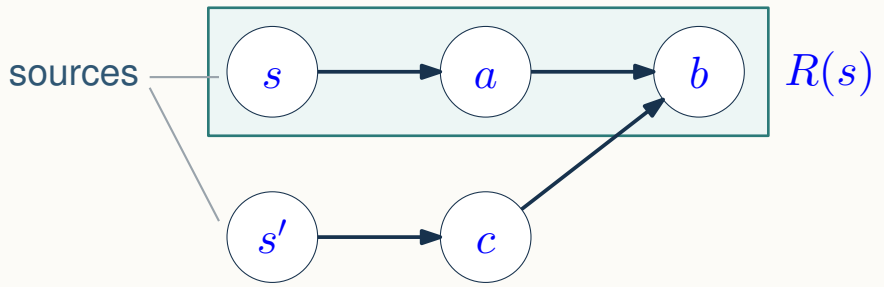
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$\Rightarrow$ we may assume that G_X is **acyclic**: every agent is reachable from a source.
$R(s)$: agents reachable from s (including s).



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	a	b	c	$v_i(X_i)$
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$S = \{a, b, c\}$, additive values

Example. $S = \{a, b, c\}$ is envied by both agents $(5 > 3,\ 5 > 2)$.

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$\{a, c\}$ is envied, but not minimal: $\{c\}$ is envied. Minimal envied subsets need not be unique.

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If Z is given to some agent and no value decreases, then nobody violates EFX towards Z.

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U_2 **(improving configuration)**: if there is one, execute it (defined later).

Rule U_0

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Proof. By definition of the rule; values only grow by monotonicity.

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U_1 **(take from the pool)**: if some agent envies P, let Z be a minimal envied subset of P and i a most envious agent with witness Z: $X_i \leftarrow Z$, $P \leftarrow (P \setminus Z) \cup X_i^{\text{old}}$.

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Question. Why does U_1 keep EFX?

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Only X_i changes, and $v_i(Z) > v_i(X_i)$.

- i towards X_k: $v_i(X_k \setminus \{g\}) \leq v_i(X_i) < v_i(Z)$.
- j towards Z: key lemma.
- all other comparisons are unchanged.

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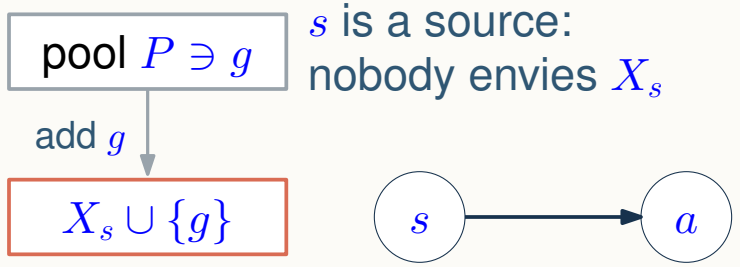
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So in the end, when U_1 does not apply, nobody envies the pool.

What If U_0 Does Not Apply?

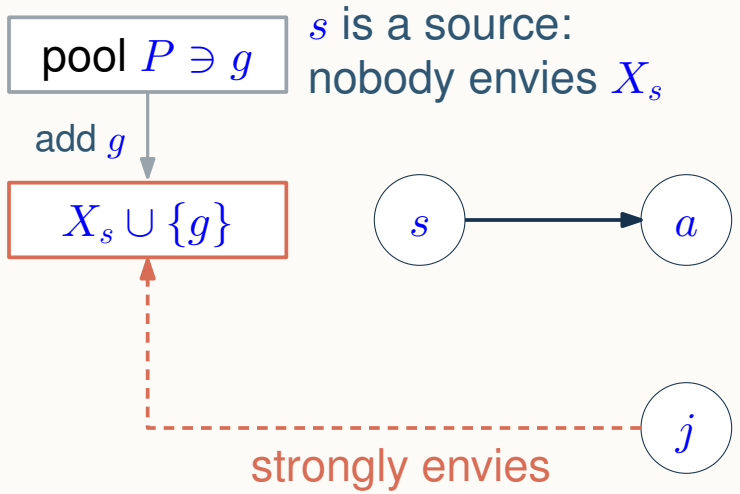
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If U_0 does not apply, the only obstacle: some j **strongly envies** $X_s \cup \{g\}$.

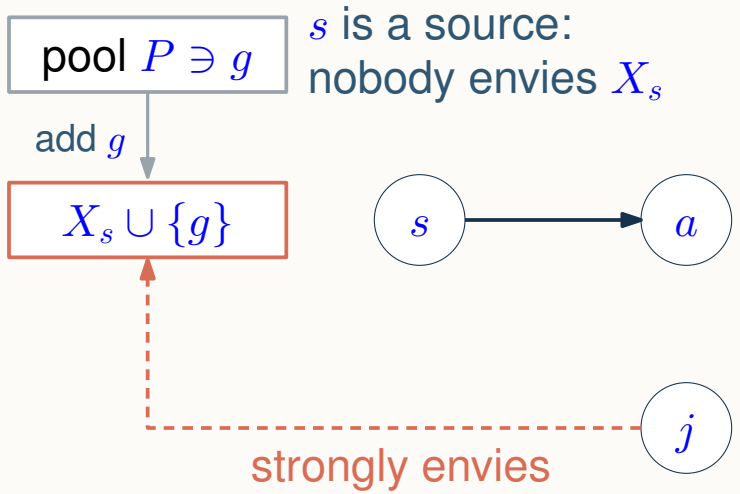


What If U_0 Does Not Apply?

Give $g \in P$ to a **source** s: s is happy, and nothing changes between the others.

If U_0 does not apply, the only obstacle: some j **strongly envies** $X_s \cup \{g\}$.

Observation. If U_0 does not apply and s is a source, then for every $g \in P$, the set $X_s \cup \{g\}$ is envied by some agent $j \neq s$.



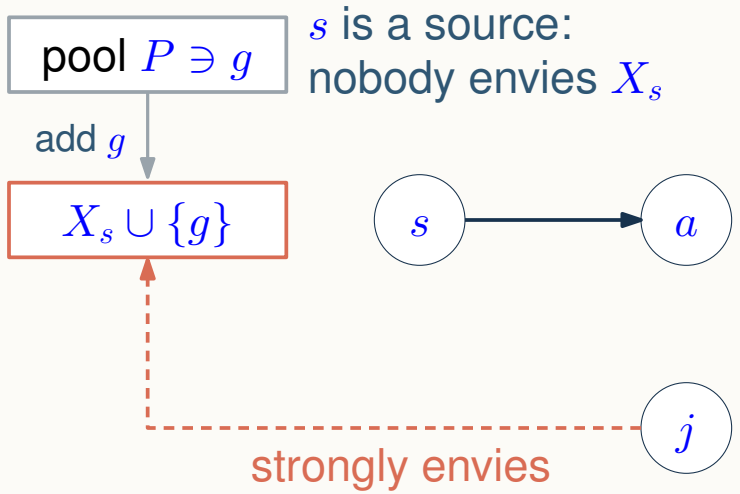
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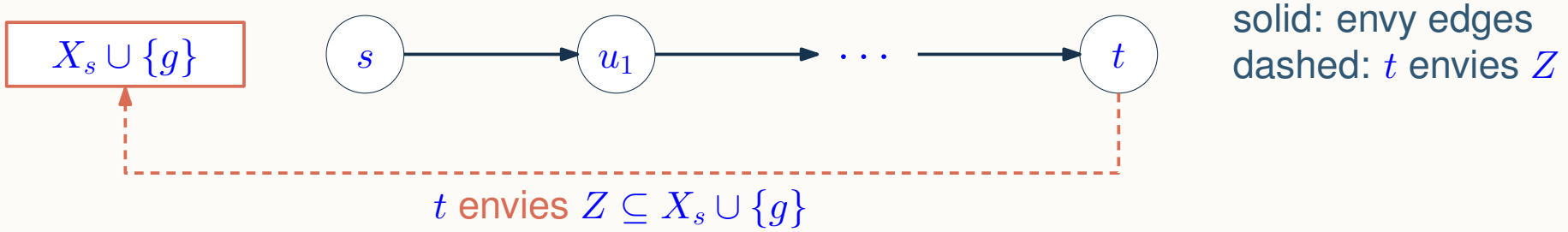
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$\Rightarrow X_s \cup \{g\}$ has a **most envious agent** t, with witness Z.



Case 1: t Is Reachable from s

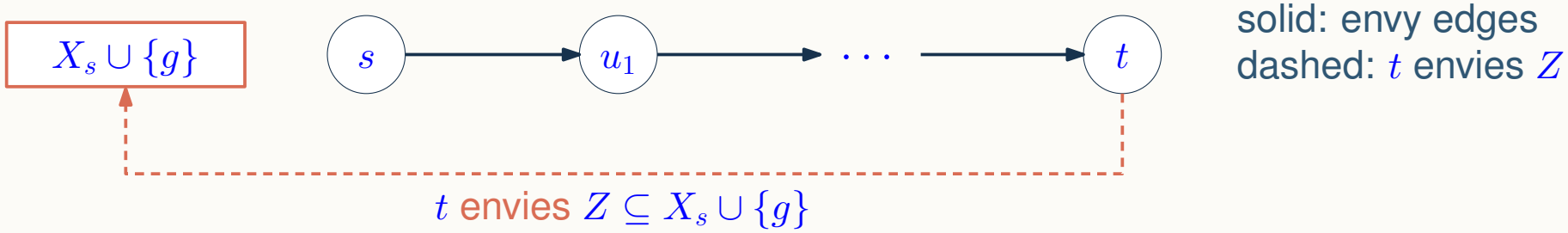
The most envious agent t of $X_s \cup \{g\}$ (witness Z) lies in $R(s)$: take a path $s = u_0 \to u_1 \to \cdots \to u_q = t$.



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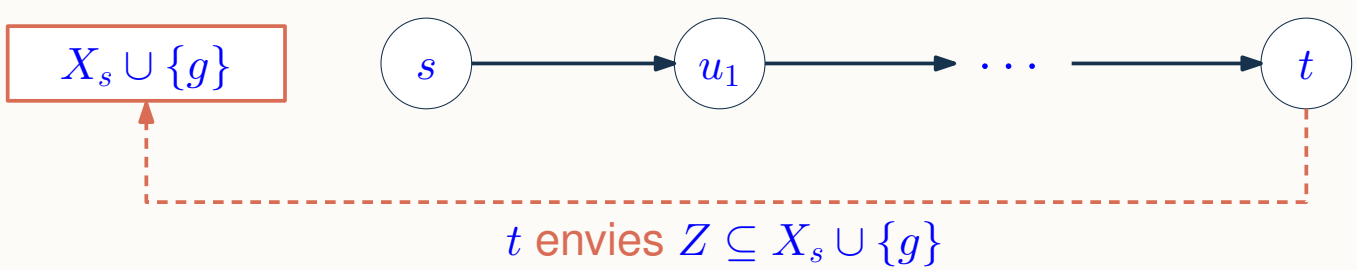


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Question. This is a cycle. What can we do?

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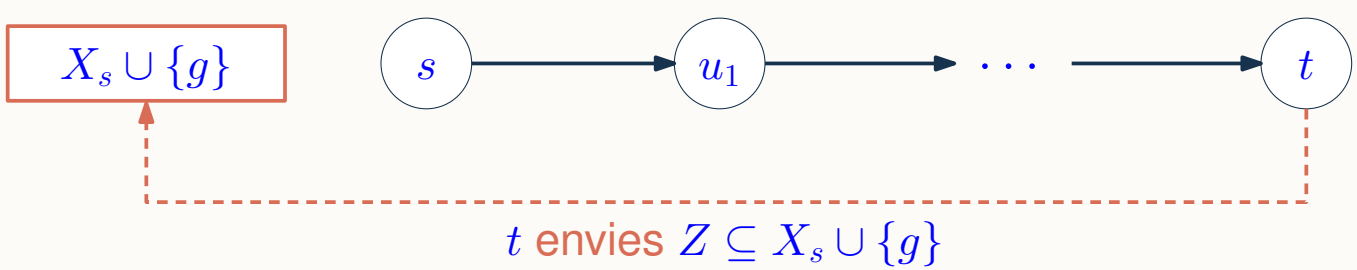


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Rotate: u_p takes $X_{u_{p+1}}$ $(p < q)$, t takes Z, the rest of $X_s \cup \{g\}$ goes to the pool.

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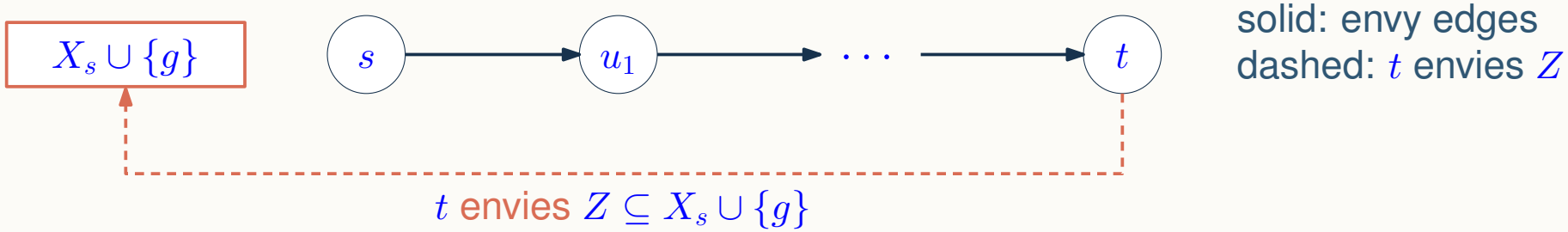
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Question. Is the new allocation EFX?

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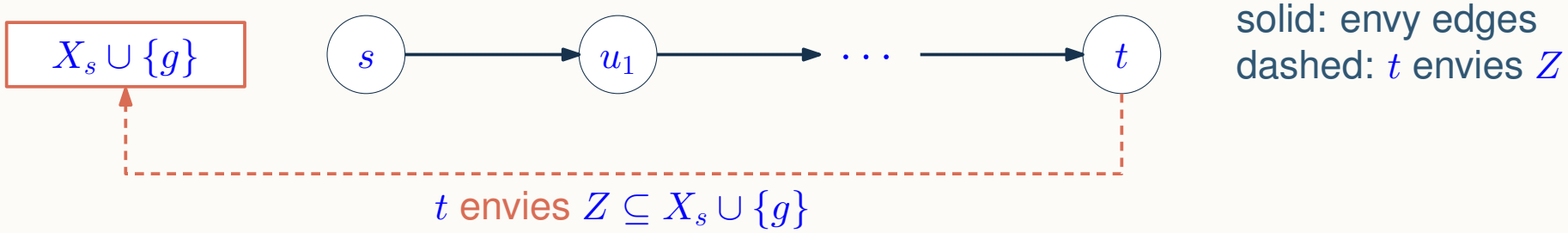
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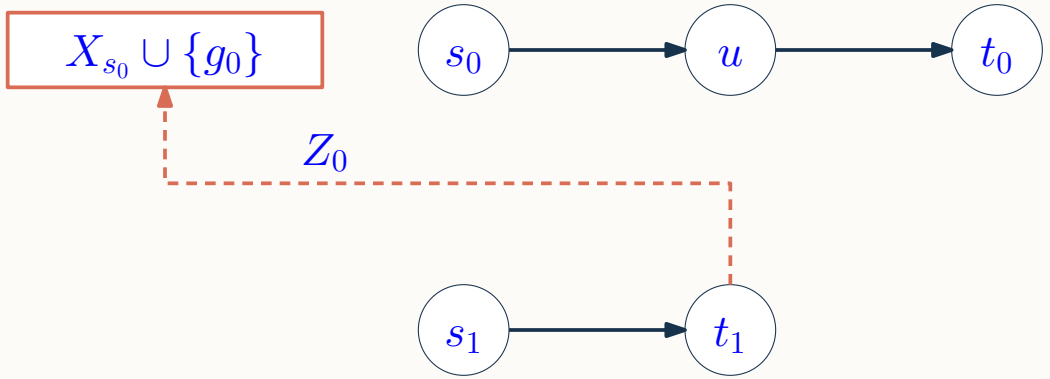
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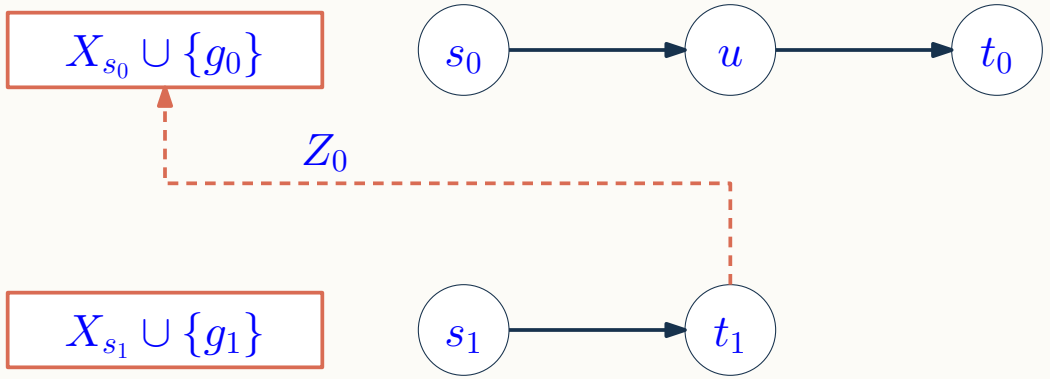
If $t = s$ $(q = 0)$: s simply replaces X_s by Z.

Case 2: t Is Not Reachable from s



t_1 (most envious agent of $X_{s_0} \cup \{g_0\}$) is not in $R(s_0)$.

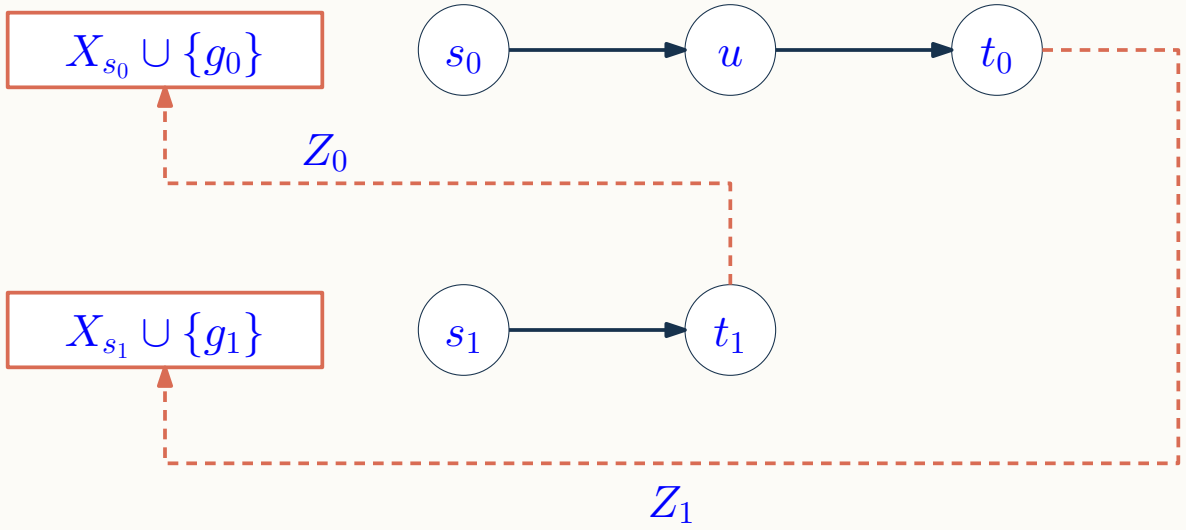
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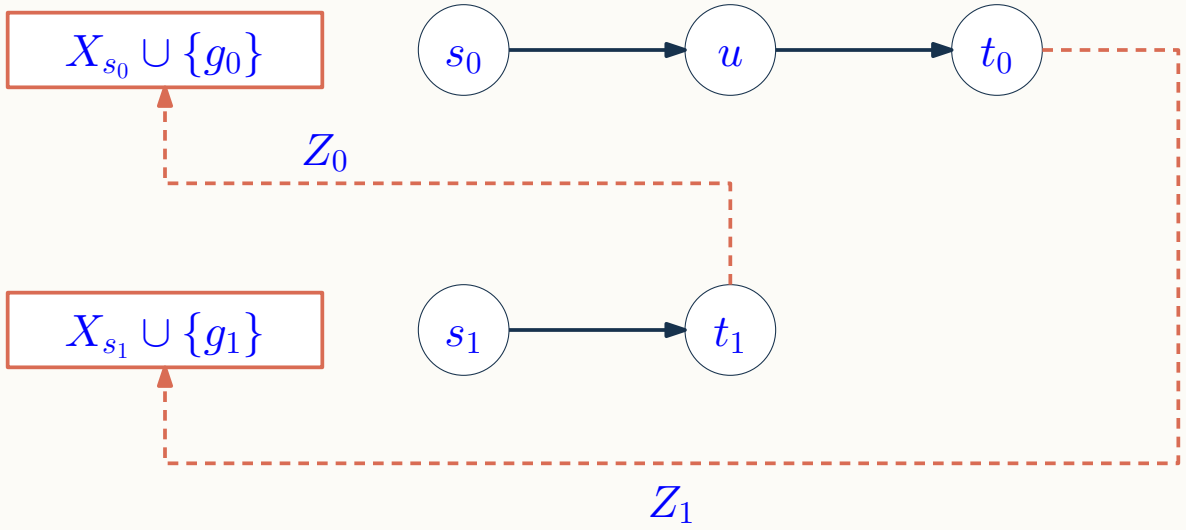


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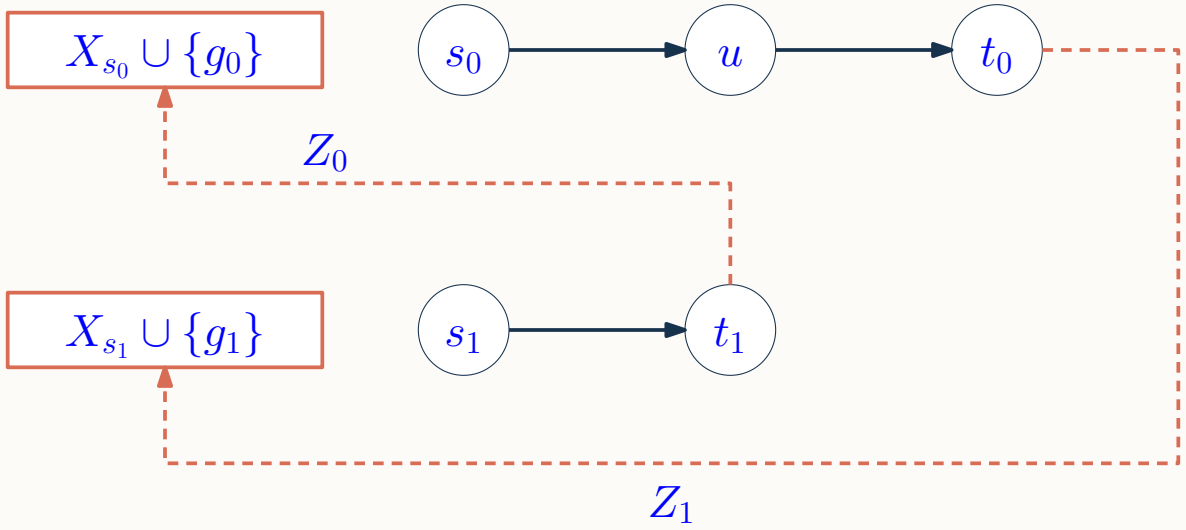
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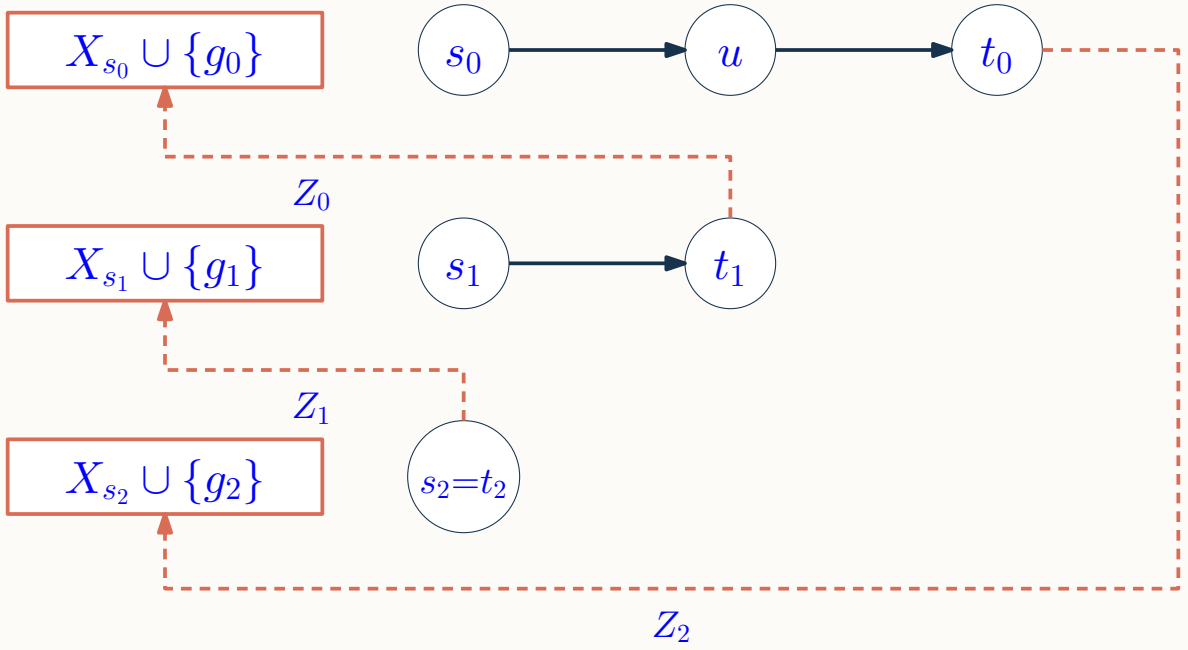
Otherwise a third source joins, and so on. Each round uses a **new** source and a **new** pool good $\Rightarrow$ a cycle closes if $|P| \geq$ number of sources.

Improving Configurations

Definition. An **improving configuration** consists of distinct sources $s_0, \ldots, s_{\ell-1}$, distinct goods $g_0, \ldots, g_{\ell-1} \in P$, and agents $t_0, \ldots, t_{\ell-1}$ such that (indices mod ℓ):

- t_{r+1} is a most envious agent of $X_{s_r} \cup \{g_r\}$, with witness Z_r;
- there are vertex-disjoint paths π_r from s_r to t_r in G_X (length 0 allowed).

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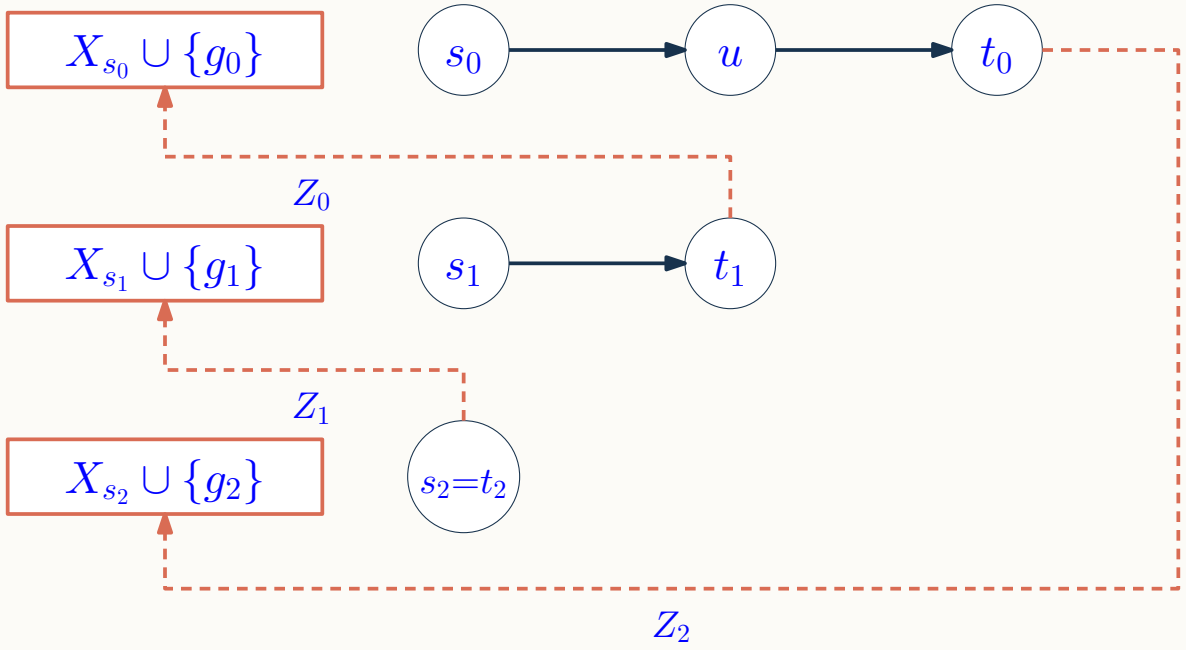


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Executing ($\ell = 3$): s_0 takes X_u, u takes X_{t_0}, t_0 takes Z_2; s_1 takes X_{t_1}, t_1 takes Z_0; $s_2 = t_2$ takes Z_1.

Case 1: $\ell = 1$. Case 2: $\ell = 2$.

Executing Keeps EFX

Lemma 2 (key lemma).
Let Z be a minimal envied subset. Then $v_j(Z \setminus \{g\}) \leq v_j(X_j)$ for all agents j and all $g \in Z$: nobody strongly envies Z.

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Executing an improving configuration keeps X EFX, weakly increases every value, and strictly increases the values of all agents on the paths.

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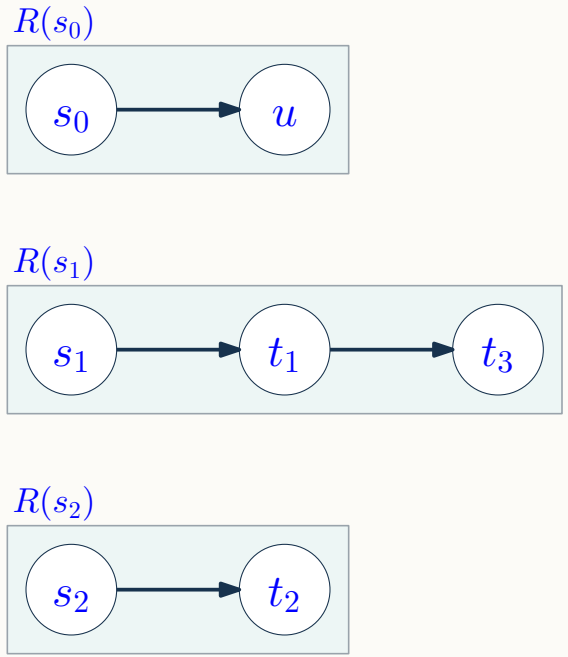
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Large Pools Admit an Improving Configuration

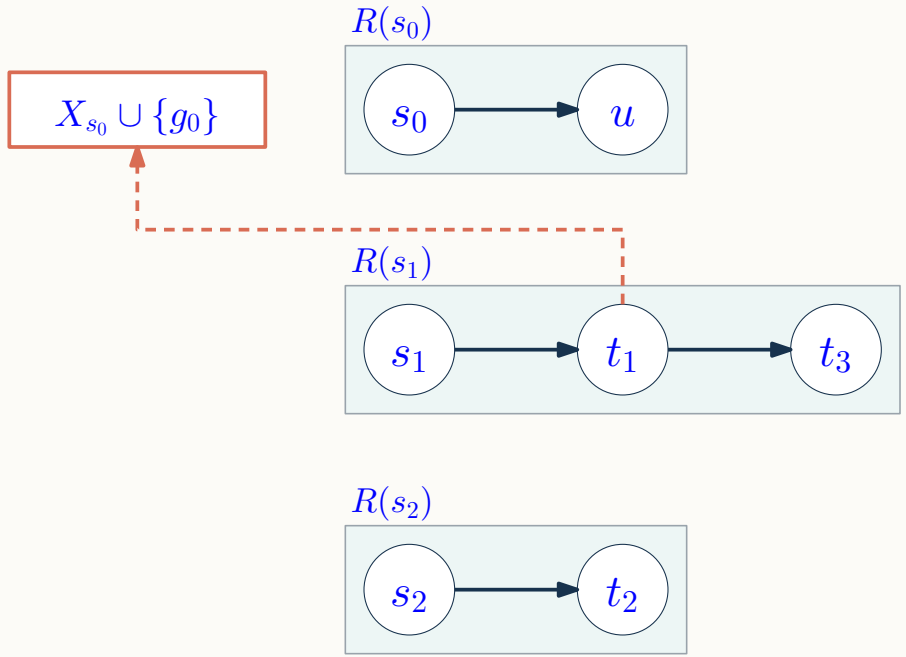
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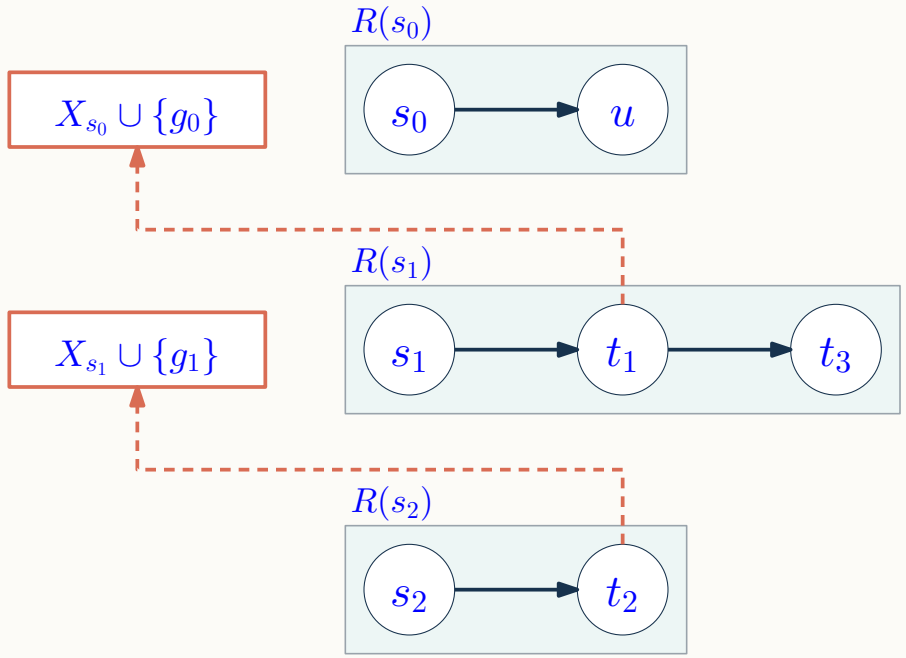


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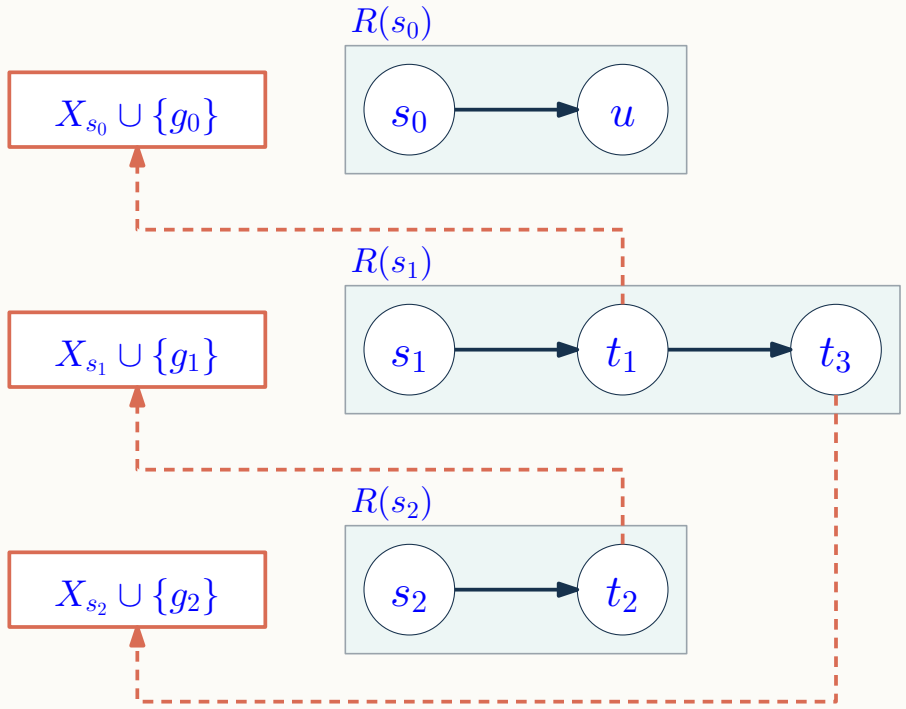
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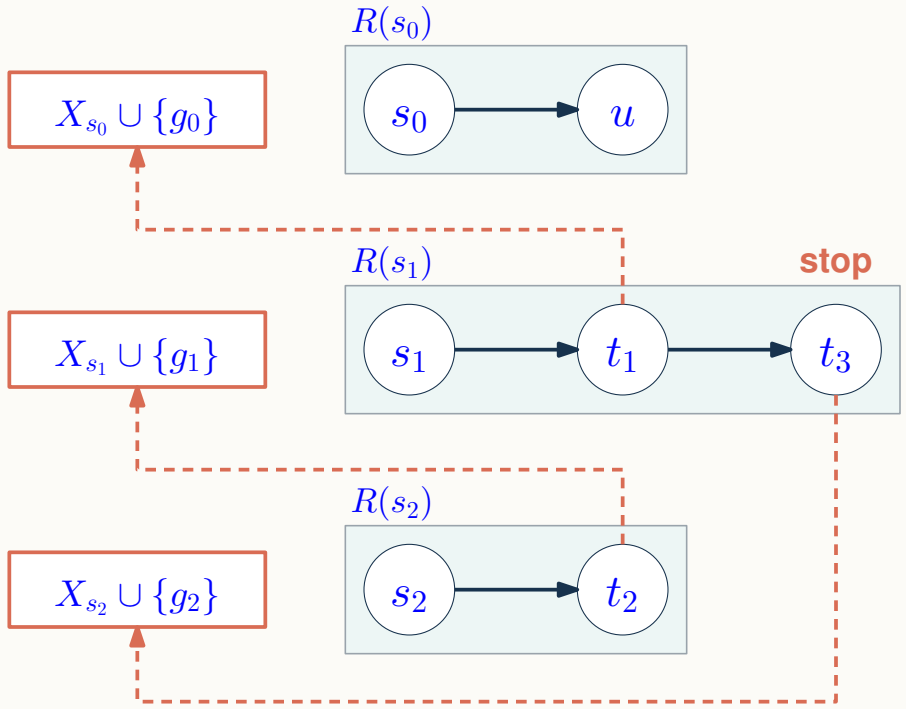
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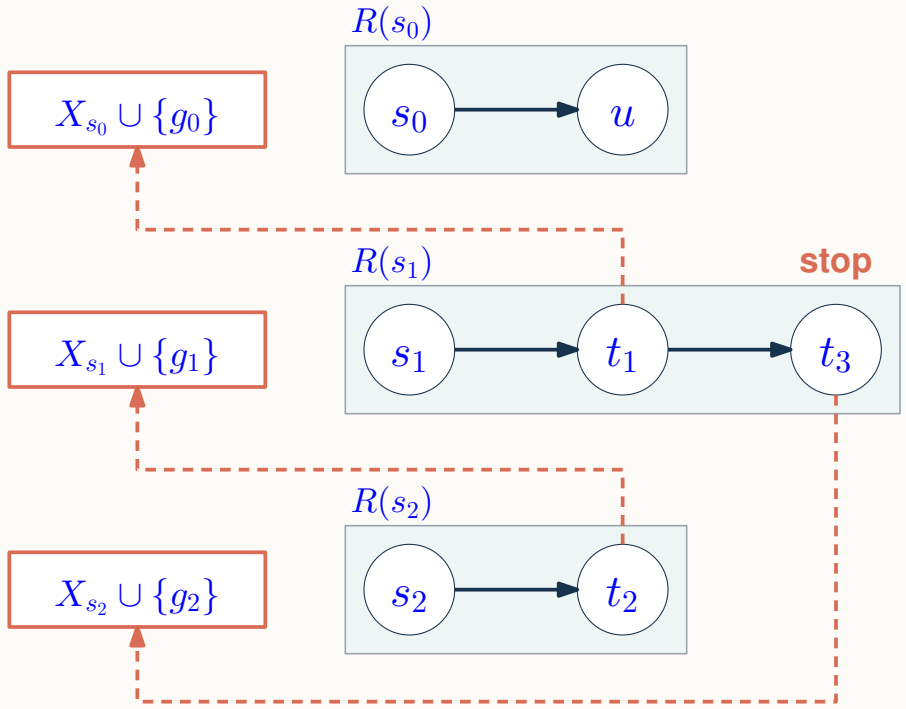
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(5) The paths are disjoint. Each $\pi_{r'}$ lies in $R(s_{r'})$. If π_r with $r > r'$ had a vertex $u \in R(s_{r'})$, then t_r (reachable from u along π_r) would be in $R(s_{r'})$, and we would have stopped in step r.



Proof of the Theorem

Theorem (Chaudhury, Kavitha, Mehlhorn, Sgouritsa 2020).
For monotone valuations, there is a partial allocation X with pool P such that (i) X is EFX, (ii) nobody envies P, (iii) $|P| <$ number of sources of $G_X \leq n$.

Lemmas 1, 3, 4, 5.
Rotations, U_0, U_1, U_2 keep X EFX and make nobody worse off. Rotations, U_1, U_2 strictly increase $\sum_i v_i(X_i)$.

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Termination: every step increases $\left(\sum_i v_i(X_i),\ -|P|\right)$ lexicographically (U_0 decreases $|P|$); finitely many allocations.

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For monotone valuations, there is a partial allocation X with pool P such that (i) X is EFX, (ii) nobody envies P, (iii) $|P| <$ number of sources of $G_X \leq n$.

Lemmas 1, 3, 4, 5.
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Lemma 6.
If U_0 does not apply, G_X is acyclic and $|P| \geq$ number of sources, then there is an improving configuration.

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Termination: every step increases $\left(\sum_i v_i(X_i),\ -|P|\right)$ lexicographically (U_0 decreases $|P|$); finitely many allocations.

Question. Why does the output satisfy (i)–(iii)?

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Output: (i) invariant; (ii) U_1 does not apply; (iii) U_0 and U_2 do not apply $\Rightarrow$ by Lemma 6, $|P| <$ number of sources.

Charity and Nash Welfare

Observation. No rotation and no rule makes any agent worse off. $\Rightarrow$ We may start from **any** EFX partial allocation X^0, and the output has $v_i(X_i) \geq v_i(X_i^0)$ for all i.

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For additive valuations and an MNW allocation X^*, there is an EFX partial allocation Y with $v_i(Y_i) \geq \frac{1}{2} v_i(X_i^*)$ for all i, and $Y_{i_0} = X_{i_0}^*$ for some i_0.

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Start the algorithm from Y: $v_i(X_i) \geq v_i(Y_i)$, so $\mathrm{NW}(X)^n \geq \prod_i v_i(Y_i) \geq 2^{-(n-1)} \operatorname{opt}^n$.

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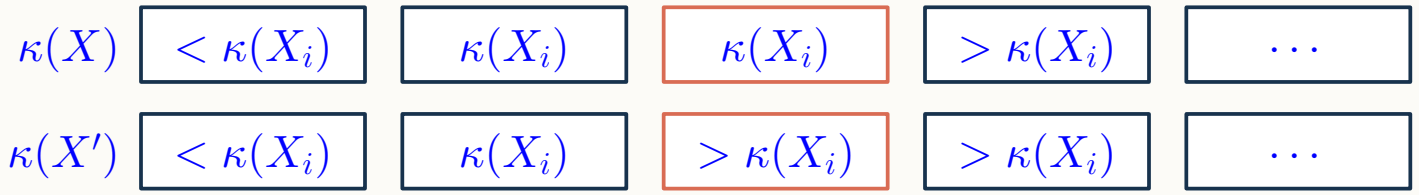
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first difference

Question. Why $|B|$?

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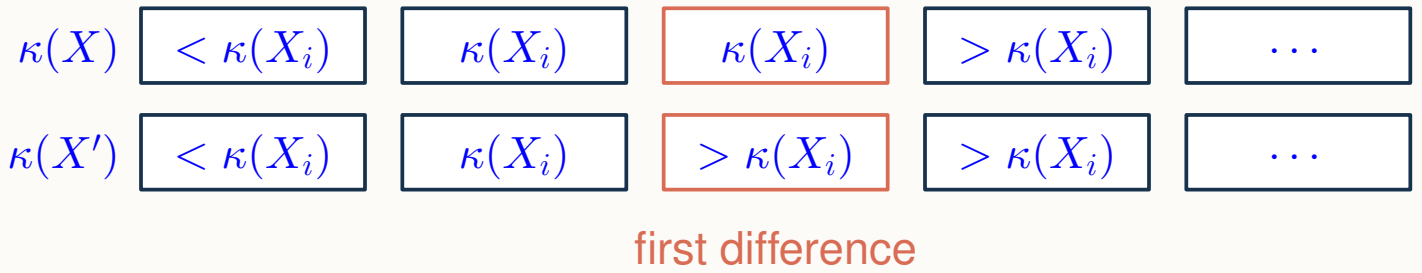
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If $v(X_i \cup \{g\}) = v(X_i)$, no value changes; the size breaks the tie.

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For $n \geq 3$ agents with different monotone valuations, EFX allocations need not exist.

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- Monotone valuations: some charity is unavoidable. Additive valuations: is $P = \emptyset$ always possible? **Open**.

Summary

Partial EFX	Donated goods	Envy towards pool	Nash welfare
Caragiannis, Gravin, Huang 2019	possibly many	possible	$\geq 2^{-(1-1/n)}\,\mathrm{opt}$
Chaudhury, Kavitha, Mehlhorn, Sgouritsa 2020	$< n$	none	$\geq 2^{-(1-1/n)}\,\mathrm{opt}$

Nash welfare guarantees for additive valuations; the charity result holds for monotone valuations.

Exercises

Q 1 (Identical valuations). Suppose all agents have the same monotone valuation v. Show that whenever $P \neq \emptyset$, one of the rules U_0, U_1, U_2 applies. Conclude that a complete EFX allocation exists (another proof of the Plaut–Roughgarden theorem).

Q 2 (Computing a minimal envied subset). Let S be envied. Go through the goods of S in any order, and remove the current good whenever the remaining set is still envied by some agent. Show that the result is a minimal envied subset of S, and that $O(n|S|)$ value queries suffice.

Q 3 (Running time). Consider the variant in which U_2 is applied only if U_0 does not apply and $|P|$ is at least the number of sources, using the configuration from the proof of Lemma 6.
(a) Show that this variant still proves the theorem.
(b) Suppose all values are integers and $V = \max_i v_i(M)$. Show that the variant performs at most $(nV + 1)(m + 1)$ applications of U_0, U_1 and U_2, each with polynomially many value queries.